



**EAST TEXAS A&M**  
— UNIVERSITY —

# CSCI 352.61E

## DIGITAL FORENSICS

# ADVANCED SEARCH

# TOPICS

- Search information inside of folder
- Search information inside of a DD image

SEARCH INFORMATION INSIDE  
OF FOLDER

# DOWNLOAD LAB FILES

```
(student@kalit150)-[~/regex]  
$ wget -q wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/regex/regex.zip  
  
(student@kalit150)-[~/regex]  
$ ls -l regex.zip  
-rw-r--r-- 1 student student 3288126 Apr 22 22:37 regex.zip
```

4 x

unzip the file

```
(student@kalit150)-[~/regex]
$ unzip regex.zip
Archive:  regex.zip
  inflating: 0_Number_Systems.pptx
  inflating: active_contracts_by_vendor_name_excluded.xls
  inflating: Animation.html
  inflating: customers.docx
  inflating: CYFI727_Roster_2028.xlsx
  inflating: DefaultDDMStructureHelperImpl.java
  inflating: Detailed-System-Report.html
  inflating: drive_fs.txt
  inflating: informant.ost
  inflating: iolang.py
  inflating: ScriptablePointer.java
```

Verify some unzipped files

```
(student@kalit150)-[~/regex]
$ ls CYFI727_Roster_2028.* -l
-rw-r--r-- 1 student student 2623 Apr 22 23:49 CYFI727_Roster_2028.csv
-rw-r--r-- 1 student student 2623 Apr 22 23:50 CYFI727_Roster_2028.txt
-rw-r--r-- 1 student student 13202 Apr 22 21:34 CYFI727_Roster_2028.xlsx
```

List all file types in the directory, including all sub-

```
(student@kalit150)-[~/regex]  
$ ls -R | grep -Eio "\.[a-zA-Z]{3,4}\b" | sort | uniq -c | sort -nr  
3 .pdf  
3 .jpg  
2 .zip  
2 .txt  
2 .java  
2 .html  
1 .xlsx  
1 .xls  
1 .pptx  
1 .ost  
1 .docx  
1 .csv
```

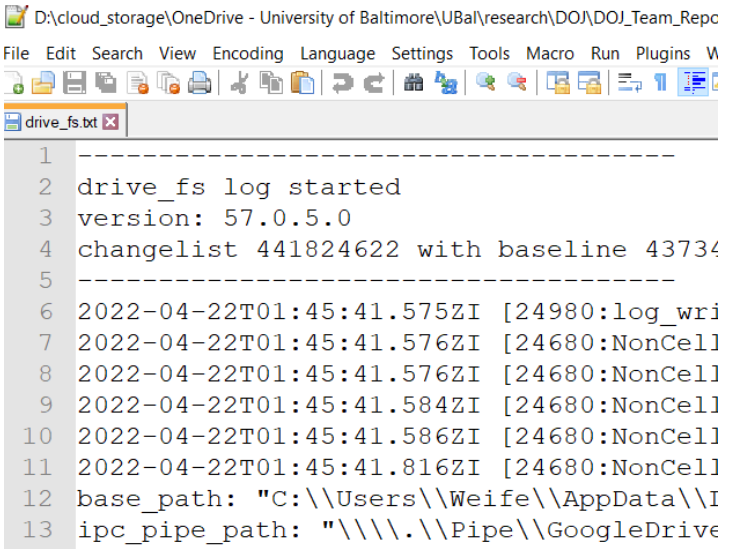
# FILE TYPES

|          | File Type          | File Eextension               | Tools                   |
|----------|--------------------|-------------------------------|-------------------------|
| unzipped | text               | .txt, .html, .java, .cpp, xml | grep                    |
|          | pdf                | .pdf                          | pdfgrep                 |
|          | database           | .db                           | Convert to text/grep -a |
|          | Outlook email      | .ost                          | pffexport               |
|          | Photo              | .jpg                          | OCR                     |
|          | Video              | .mp4                          | OCR                     |
|          | audio              | .wav                          | Deepspeech, htk         |
|          | executable         | .exe, .com, dll               | grep -a                 |
| zipped   | Compressed         | .zip, .7z                     | unzip                   |
|          | proprietary format | .docx, .xlsx, pptx            | Unzip ,<br>search xml   |



# CONTENT TYPES

- ASCII code
  - human readable
  - can be viewed in **Notepad** or **cat** command
  - e.g., .txt, Java, HTML
- Binary
  - not human readable
  - need special software to view the content
  - executable, special documents
  - .exe, .com, .jpg, .pdf
  - strings are readable



The screenshot shows a Notepad window with the title bar 'D:\cloud\_storage\OneDrive - University of Baltimore\UBal\research\DOJ\DOJ\_Team\_Rep...'. The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, and Plugins. The toolbar contains various icons for file operations. The text area shows the following content:

```
1 -----
2 drive_fs log started
3 version: 57.0.5.0
4 changelist 441824622 with baseline 43734
5 -----
6 2022-04-22T01:45:41.575ZI [24980:log_wri
7 2022-04-22T01:45:41.576ZI [24680:NonCell
8 2022-04-22T01:45:41.576ZI [24680:NonCell
9 2022-04-22T01:45:41.584ZI [24680:NonCell
10 2022-04-22T01:45:41.586ZI [24680:NonCell
11 2022-04-22T01:45:41.816ZI [24680:NonCell
12 base_path: "C:\\Users\\Weife\\AppData\\I
13 ipc_pipe_path: "\\.\Pipe\\GoogleDrive
```

Save strings of a jpg to a txt file

```
(student@kalit150)-[~/regex]
```

```
$ strings RisetothetheChallengeSaveDate1963.jpg | tr '\n' ' ' > risestrings.txt
```

```
(student@kalit150)-[~/regex]
```

```
$ more risestrings.txt
```

```
Exit Adobe Photoshop 22.1 (Windows) 2022:02:21 13:27:47 Adobe_CM Adobe b34r 7GWg  
w AQaq" dEU6te '7GWgw U:-} =\Z1_c) 70e? S]TX 7cI. h6Ye [r+q NIllC rX}2 [2  
vC +hhq sAsX h&=G ;=6o jslw y}?' {=d. I2u+ +4By?Y9Tv v^?^~s k+}o =?*9 RGF=8 z"I(  
8Photoshop 3.0 8BIM 8BIM printOutput PstSbool Intenum Inte Clrm printSixteenBi  
tbool printerNameTEXT printProofSetupObjc proofSetup Bltnenum builtinProof  
proofCMYK 8BIM printOutputOptions Cptnbool Clbrbool RgsMbool CrnCbool CntCbool L  
blsbool Ngtvbool EmlDbool Intrbool BckgObjc RGBC Rd doub@o Grn doub@o Bl doub@  
o BrdTUnf#Rlt Bld Unf#Rlt RsltUnf#Pxl@k vectorDatabool PgPsenum PgPs PgPC Lef
```

## Install a tool

```
$ sudo apt install -y exiftool
[sudo] password for kali:
Reading package lists... Done
```

## Show metadata

```
$ exiftool RisettotheChallengeSaveDate1963.jpg
ExifTool Version Number      : 12.41
File Name                    : RisettotheChallengeSaveDate1963.jpg
Directory                    : .
File Size                    : 50 KiB
File Modification Date/Time   : 2022:04:22 23:24:26-04:00
File Access Date/Time        : 2022:06:03 21:08:46-04:00
File Inode Change Date/Time   : 2022:06:02 14:57:43-04:00
File Permissions              : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Big-endian (Motorola, MM)
Orientation                  : Horizontal (normal)
X Resolution                  : 216
Y Resolution                  : 216
Resolution Unit               : inches
Software                     : Adobe Photoshop 22.1 (Windows)
Modify Date                   : 2022:02:21 13:27:47
Color Space                   : sRGB
Exif Image Width              : 196
Exif Image Height             : 196
Compression                   : JPEG (old-style)
Thumbnail Offset              : 318
Thumbnail Length              : 2786
IPTC Digest                   : 00000000000000000000000000000000
```

## Search email in all text files

```
(student@kalit150)-[~/regex]
$ find . -type f -name "*.txt" -print -exec grep -Eo "[a-zA-Z0-9._]+@[a-zA-Z0-9._]+\.[a-zA-Z0-9]{3}" {} \;
```

end of the command.  
'\ ' is an escape symbol

placeholder for each matched file

./CYFI727\_Roster\_2028.txt  
cjonathan@gmail.com  
crichard@hotmail.com  
davis.richard@ubalt.edu  
german\_pau001@gmail.com  
david@md.gov  
kevin001@mit.edu  
mark.jacobson@gannon.edu  
mckyle\_cool@example.com  
frank.xu@gmail.com  
ziebell.jonathan@rsa.org  
./drive\_fs.txt  
frank.w.xu@gmail.com

## Search email in pdf files

```
(student@kalit150)-[~/regex]
$ find . -iname '*.pdf' -exec pdftotext -Po "[a-zA-Z0-9._]+@[a-zA-Z0-9._]+\.[a-zA-Z0-9]{3}" {} +
./sample_resumes.pdf:careerservices@bellevue.edu
./sample_resumes.pdf:imasample1@xxx.com
./sample_resumes.pdf:imasample2@xxx.com
./sample_resumes.pdf:imasample3@xxx.com
./sample_resumes.pdf:imasample4@xxx.com
./sample_resumes.pdf:imasample5@xxx.com
./sample_resumes.pdf:imasample6@xxx.com
./sample_resumes.pdf:imasample7@xxx.com
./sample_resumes.pdf:imasample8@xxx.com
./sample_resumes.pdf:imasample9@xxx.com
./sample_resumes.pdf:imasample10@xxxx.net
./sample_resume.pdf:jwsmith@colostate.edu
```

```
sudo apt-get install -y pdftotext
```

### Functional Resume Sample

#### John W. Smith

2002 Front Range Way Fort Collins, CO 80525

[jwsmith@colostate.edu](mailto:jwsmith@colostate.edu)

#### Career Summary

Four years experience in early childhood development with a diverse background in the care of special needs children and adults.

#### Adult Care Experience

- Determined work placement for 150 special needs adult clients.
- Maintained client databases and records.
- Coordinated client contact with local health care professionals on a monthly basis.
- Managed 25 volunteer workers.

#### Childcare Experience

- Coordinated service assignments for 20 part-time counselors and 100 client families.
- Oversaw daily activity and outing planning for 100 clients.
- Assisted families of special needs clients with researching financial assistance and healthcare.
- Assisted teachers with managing daily classroom activities.
- Oversaw daily and special student activities.



Show .docx content without unzip to disk

-p extract files to pipe, no messages

```
(student@kalit150)-[~/regex]
```

```
$ unzip -p "*.docx" word/document.xml | sed -E 's/<[^<]*paraId[^>]*>/ /g ; s/<[^>]+>/ /g'
```

Most cybersecurity companies, such as Palo Alto Networks, Cisco, Trend Micro, focus on stopping real-time cyberattacks. Our company provides services before and after attacks. We provide customized evidence collecting plans and focus on reconstructing cybercrime scenarios and identity attackers based on the collected digital evidence from computing devices, operating systems, application logs, and networks. Also, we are different from these commercial digital forensic software companies that provide forensic tools, such as x-way, Encase, Magnet: we focus on formalizing, storing, and visualizing evidence in knowledge graphs. We utilize knowledge graphs to discover evidence, validate evidence, and reconstruct crime scenarios with traceability functions. Frank Xu wxu@ubalt.edu

Show .docx props without unzip to disk

```
(student@kalit150)-[~/regex]
```

```
$ unzip -p "*.docx" docProps/core.xml | sed -E 's/<[^>]+>/ /g'
```

Weifeng Xu

Weifeng Xu

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2022-04-22T14:47:00Z

2022-04-23T01:42:00Z

Show email in a .docx content without unzip to disk

```
(student@kalit150)-[~/regex]  
$ unzip -p "*.docx" word/document.xml | sed -E 's/<[^<]*paraId[^>]*>/ /g ; s/<[^>]+>//g'  
| grep -Eo "[a-zA-Z0-9._]+@[a-zA-Z0-9._]+\.[a-zA-Z0-9]{3}"
```

wxu@ubalt.edu

Try multiple docx files

```
(student@kalit150)-[~/regex]  
$ cp customers.docx cu.docx
```



Only can search word/document.xml

```
(student@kalit150)-[~/regex]  
$ find . -type f -name "*.docx" -print -exec unzip -p {} word/document.xml \; | sed -E  
's/<[^<]*paraId[^>]*>/ /g ; s/<\\w:document>/\\n/g; s/<[^>]+>//g'
```

./customers.docx

insert a new line between files

Most cybersecurity companies, such as Palo Alto Networks, Cisco, Trend Micro, focus on stopping real-time cyberattacks. Our company provides services before and after attacks. We provide customized evidence collecting plans and focus on reconstructing cybercrime scenarios and identity attackers based on the collected digital evidence from computing devices, operating systems, application logs, and networks. Also, we are different from these commercial digital forensic software companies that provide forensic tools, such as x-way, Encase, Magnet. we focus on formalizing, storing, and visualizing evidence in knowledge graphs. We utilize knowledge graphs to discover evidence, validate evidence, and reconstruct crime scenarios with traceability functions. Frank Xu wxu@ubalt.edu

./cu.docx

Most cybersecurity companies, such as Palo Alto Networks, Cisco, Trend Micro, focus on stopping real-time cyberattacks. Our company provides services before and after attacks. We provide customized evidence collecting plans and focus on reconstructing cybercrime scenarios and identity attackers based on the collected digital evidence from computing devices, operating systems, application logs, and networks. Also, we are different from these commercial digital forensic software companies that provide forensic tools, such as x-way, Encase, Magnet: we focus on formalizing, storing, and visualizing evidence in knowledge graphs. We utilize knowledge graphs to discover evidence, validate evidence, and reconstruct crime scenarios with traceability functions. Frank Xu wxu@ubalt.edu



Only can search docProps/core.xml

```
(student@kalit150)-[~/regex]
$ find . -type f -name "*.docx" -print -exec unzip -p {} docProps/core.xml \; | sed -E
's/<\/cp:coreProperties>/\n/g;s/<[^>]+>/ /g;'
./customers.docx
    Weifeng Xu    Weifeng Xu    53    2022-04-22T14:47:00Z    2022-04-23T01:42:00Z
./cu.docx
    Weifeng Xu    Weifeng Xu    53    2022-04-22T14:47:00Z    2022-04-23T01:42:00Z
```

add a new line between files

Search people's names in images (mainly header or hidden names)

use space " "

```
(student@kalit150)-[~/regex]
$ find . -type f -name "*.jpg" -print -exec grep -Eoa "\b[A-Z][a-z]{1,10}\s+[A-Z][a-z]{1,10}\b" {} \;
./RisetothetheChallengeSaveDate1963.jpg
Adobe Photoshop
Adobe Photoshop
Adobe Photoshop
Adobe Photoshop
Packard Company
Reference Viewing
Reference Viewing
./ub_building2.jpg
./ub_building.jpg
```

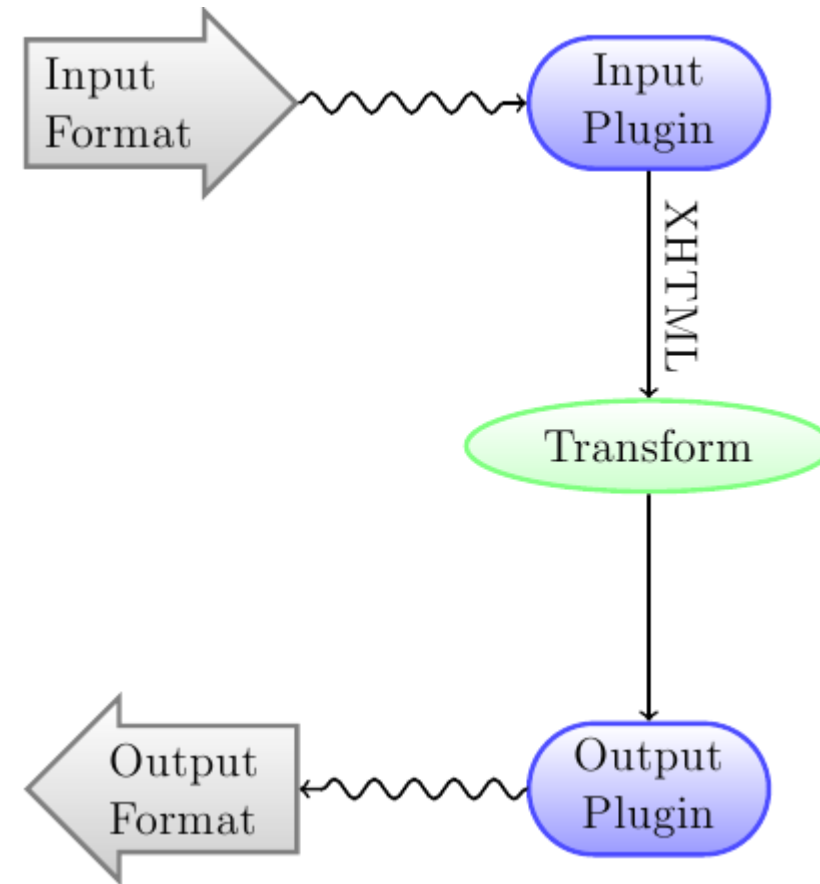
People's names have patterns,  
however, the results are inaccurate

# **CALIBRE** CONVERSION

# CALIBRE

```
$ ebook-convert --help
Usage: ebook-convert input_file output_file [options]
```

- Convert an e-book from one format to another.
- The output e-book format is guessed from the file extension of output\_file.
- We use the software to convert files with different format to text file.
  - supported: docx, pdf
  - **not** supported: xlsx, pptx



Install calibre to convert pdf to

```
text
(kali@kali)-[~/regex]
└─$ sudo -v && wget -nv -O- https://download.calibre-ebook.com/linux-installer.sh | sudo sh /dev/stdin
Using python executable: /usr/bin/python3
2022-06-04 08:14:49 URL:https://download.calibre-ebook.com/linux-installer.sh [31922/31922] -> "-" [1]
Installing to /opt/calibre
Downloading tarball signature securely...
Will download and install calibre-5.43.0-x86_64.txz
                                     Downloading calibre-5.43.0-x86_64.txz
14% [=====-----]
```

convert pdf to text

```
$ ebook-convert sample_resume.pdf sample_resume.txt
1% Converting input to HTML...
InputFormatPlugin: PDF Input running
on /home/kali/regex/sample_resume.pdf
pdftohtml log:
```



## Show converted txt file

```
$ cat sample_resume.txt
```

Functional Resume Sample

John W. Smith

2002 Front Range Way Fort Collins, CO 80525

jwsmith@colostate.edu

### Career Summary

Four years experience in early childhood development with a diverse background in the care of special needs children and adults.

### Adult Care Experience

- Determined work placement for 150 special needs adult clients.
- Maintained client databases and records.
- Coordinated client contact with local health care professionals on a monthly basis.

**SEARCH DATA LEAKAGE IMAGE**

# HOW TO DOWNLOAD THE DD IMAGE?

Download DD images. See notes if the link expired

```
(kali@kali)-[~/lab]
$ wget https://cfreds-archive.nist.gov/data_leakage_case/images/pc/cfreds_2015_data_leakage_pc.7z.001
--2022-06-02 07:42:03-- https://cfreds-archive.nist.gov/data_leakage_case/images/pc/cfreds_2015_data_leakage_pc.7z.001
```

Repeat **wget** command to download **.002** and **.003**

[https://cfreds-archive.nist.gov/data\\_leakage\\_case/images/pc/cfreds\\_2015\\_data\\_leakage\\_pc.7z.001](https://cfreds-archive.nist.gov/data_leakage_case/images/pc/cfreds_2015_data_leakage_pc.7z.001)

[https://cfreds-archive.nist.gov/data\\_leakage\\_case/images/pc/cfreds\\_2015\\_data\\_leakage\\_pc.7z.002](https://cfreds-archive.nist.gov/data_leakage_case/images/pc/cfreds_2015_data_leakage_pc.7z.002)

[https://cfreds-archive.nist.gov/data\\_leakage\\_case/images/pc/cfreds\\_2015\\_data\\_leakage\\_pc.7z.003](https://cfreds-archive.nist.gov/data_leakage_case/images/pc/cfreds_2015_data_leakage_pc.7z.003)

Verify three zipped files

```
root@kali:~/lab# ls -l
total 5300596
-rw-r--r-- 1 root root 2147483648 Apr 29 2015 cfreds_2015_data_leakage_pc.7z.001
-rw-r--r-- 1 root root 2147483648 Apr 29 2015 cfreds_2015_data_leakage_pc.7z.002
-rw-r--r-- 1 root root 1132827932 Apr 29 2015 cfreds_2015_data_leakage_pc.7z.003
```

unzipped the DD image

```
root@kali:~/lab# 7z e cfreds_2015_data_leakage_pc.7z.001
```

Verify the unzipped DD image

```
root@kali:~/lab# ls -l
total 26272120
-rw-r--r-- 1 root root 2147483648 Apr 29 2015 cfreds_2015_data_leakage_pc.7z.001
-rw-r--r-- 1 root root 2147483648 Apr 29 2015 cfreds_2015_data_leakage_pc.7z.002
-rw-r--r-- 1 root root 1132827932 Apr 29 2015 cfreds_2015_data_leakage_pc.7z.003
-rw-r--r-- 1 root root 21474836480 Apr 21 2015 cfreds_2015_data_leakage_pc.dd
root@kali:~/lab#
```

Verify the unzipped DD image with MD5

```
$ md5sum cfreds_2015_data_leakage_pc.dd
a49d1254c873808c58e6f1bcd60b5bde cfreds_2015_data_leakage_pc.dd
```

## Types of files

```
(kali㉿kali)-[/media/kali/C8CA0C8DCA0C7A48/Users/informant]  
$ ls -R | grep -Eio "\.[a-zA-Z]{3,4}\b" | sort | uniq -c | sort -nr
```

```
595 .jpg  
410 .gif  
406 .png  
403 .htm  
374 .json  
185 .txt  
170 .tmp  
85 .dat  
84 .css  
69 .xml  
55 .ini  
43 .lnk  
41 .ico  
33 .NET  
32 .dll  
29 .com
```

```
29 .com  
28 .min  
25 .rtf  
25 .log  
19 .url  
17 .woff  
16 .svg  
16 .eot  
12 .wpl  
12 .exe  
11 .nmf  
10 .emf  
8 .XSL  
8 .DAT  
8 .bmp  
7 .LNK  
7 .feed
```

```
6 .oab  
6 .msu  
6 .ldb  
6 .html  
6 .cvr  
6 .cab  
6 .bin  
4 .xsl  
4 .swf  
4 .sol  
4 .msi  
4 .jrs  
4 .gov  
4 .comp  
4 .cdf  
3 .old  
3 .FSD  
3 .docx  
3 .aspx
```

```
2 .yui  
2 .xps  
2 .xlsx  
2 .Word  
2 .wmf  
2 .wmdb  
2 .WMC  
2 .pptx  
2 .pat  
2 .net  
2 .mzz  
2 .MSO  
2 .msn  
2 .jpeg  
2 .etl  
2 .dotx  
2 .dotm  
2 .dgps  
2 .chk  
2 .blf  
2 .ashx
```

```
1 .zip  
1 .xsd  
1 .XML  
1 .Wol  
1 .wol  
1 .wer  
1 .wav  
1 .top  
1 .tpl  
1 .TMP  
1 .thmx  
1 .srs  
1 .snt  
1 .pbk  
1 .ost  
1 .nexe  
1 .move  
1 .ltr  
1 .load  
1 .lex
```

```
1 .inf  
1 .idx  
1 .ibm  
1 .HTM  
1 .head  
1 .FSF  
1 .ersy  
1 .DTD  
1 .DIC  
1 .data  
1 .core  
1 .bing  
1 .bak  
1 .apSD  
1 .acl
```



```
(kali㉿kali)-[/media/kali/C8CA0C8DCA0C7A48/Users/informant]
$ find . -type f -name "*.jpg" -print -exec grep -Eoa "\b[A-Z][a-z]{1,10}+[A-Z][a-z]{1,10}\b" {} \; | more
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBf0hY0[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBh0W6o[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBhHuu9[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiAyNr[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiB07V[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiBb76[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiBWww[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiBXct[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiC5ob[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiC6Bn[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBijs0f[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBijUBA[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBimjCJ[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBivlpc[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiwB07[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiwzT8[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiyEnw[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBiyswS[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBizpZr[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/AA9QA7z[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/AA9sWtD[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/AA9TlSL[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/AA9TnuR[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/BBizSjS[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/benefits-1[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/benefits-5[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/build-marquee[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/chrome-existing[1].jpg
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/cubeslam-marquee[1].jpg
Adobe Photoshop ←
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/download[1].jpg
Adobe Photoshop ←
./AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/4C689SI3/kickwithchrome[1].jpg
Adobe Photoshop ←
```



# SEARCH IOS IMAGE

see [An Introduction to iOS 13 Image](#)

Types of files based on file extension (1 - 7 chars) %f File's name with any leading directories removed

```
(kali㉿kali)-[ /.../ios_13_4_1/iOS 13.4.1 Extraction/Extraction/Volumes ]  
$ find . -type f -printf "%f\n" | grep -Eio "\.[a-zA-Z]{1,7}\b" | sort | uniq -ic | sort -nr > iosfiles.txt
```

Show first 20 lines

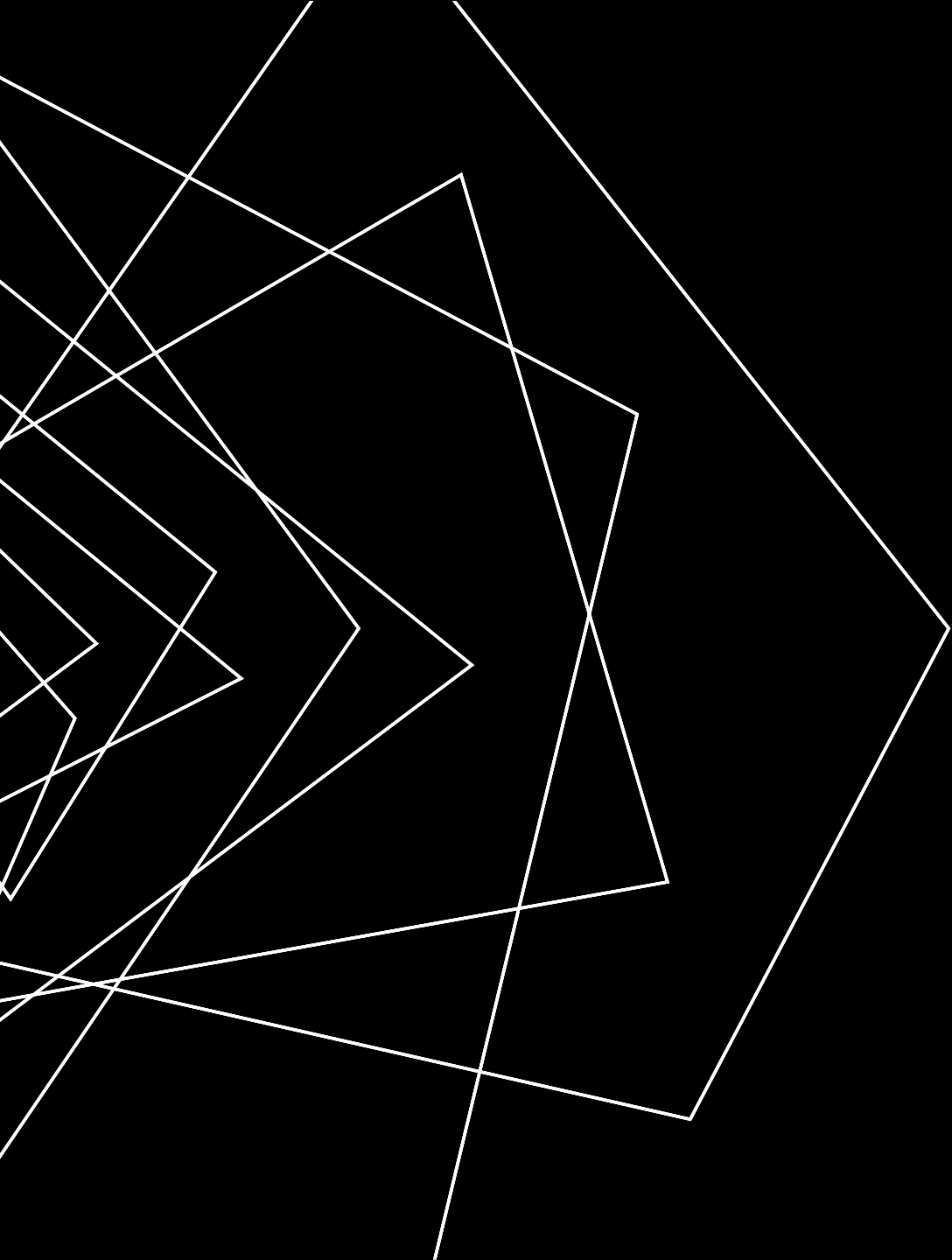
```
$ head -20 iosfiles.txt  
110824 .strings  
21661 .png  
18150 .plist  
10036 .nib  
8295 .keys  
6643 .apple  
6272 .js  
3071 .cat  
3057 .bin  
2729 .json  
1644 .com  
1402 .db  
1239 .mom  
1235 .min  
1170 .heic  
1157 .module  
1086 .jpg  
997 .xml  
980 .html  
921 .dat
```

Show first 20 -40 lines

```
$ head -40 iosfiles.txt | tail +20  
921 .dat  
839 .data  
769 .austrip  
692 .txt  
636 .car  
594 .caf  
593 .string  
522 .ips  
515 .svg  
485 .sqlite  
452 .scnz  
411 .summary  
357 .atx  
330 .astc  
325 .p  
324 .pri  
284 .caml  
279 .osddata  
274 .private  
272 .otf  
272 .btb
```

Show first 40-60 lines

```
$ head -60 iosfiles.txt | tail +40  
272 .btb  
263 .zip  
247 .ttf  
237 .maps  
224 .glsl  
224 .alloy  
220 .twimg  
220 .ktx  
219 .wam  
215 .webp  
212 .csv  
179 .uchr  
178 .log  
177 .zst  
158 .aiff  
155 .syncd  
154 .bdg  
145 .ca  
136 .css  
130 .wav  
129 .fst
```



# THANK YOU

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